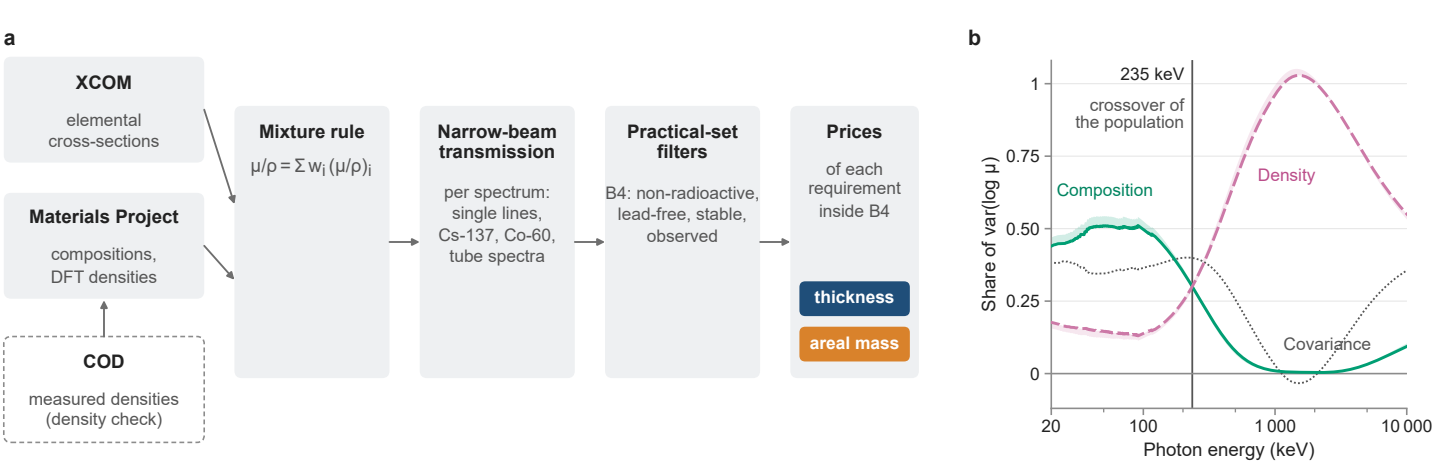

a
XCOM
elemental cross-sections
Materials Project
compositions, DFT densities
COD
measured densities (density check)
Mixture rule
μ/ρ = Σ wi (μ/ρ)i
Narrow-beam transmission
per spectrum: single lines, Cs-137, Co-60, tube spectra
Practical-set filters
B4: non-radioactive, lead-free, stable, observed
Prices
of each requirement inside B4
thickness
areal mass
b
235 keV
crossover of the population
Composition
Density
Covariance
Share of var(log μ)
1
0.75
0.50
0.25
0
20
100
1 000
10 000
Photon energy (keV)